

Noted

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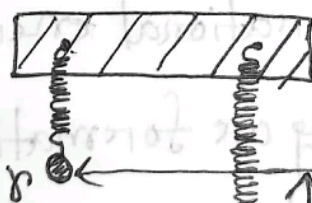
Oscillation:

* Hook's law:

$$F = -kx$$

Hook's law of vibration:

The type of vibratory motion of a body such that the restoring force in the oscillation acting on the body is directly proportional to the displacement from the mean position and always directed towards the mean position is called the simple harmonic oscillation/motion (SHO/SHM)



The force is directly proportional to the displacement but opposite in direction.

$$F \propto (-s)$$

$$F = -ks$$

k is a force constant

When the object is released, the restoring force produces an acceleration. This acceleration is proportional to the mass being accelerated.

$$\vec{F} = m\vec{a}$$

$$\vec{a} = \frac{\vec{F}}{m}$$

$$= \frac{-k\vec{s}}{m}$$

$$\vec{a} = -\left(\frac{k}{m}\right)\vec{s}$$

$$\therefore \vec{a} \propto -\vec{s}$$

So, acceleration is proportional to displacement but opposite in direction.

It's called the Hook's law

Characteristics of simple harmonic motion

1. Its motion is periodic
2. At particular time interval the motion becomes opposite.
3. It's motion is clockwise a straight line.
4. Acceleration is proportional to displacement
5. Accelerating points towards the main position of the object.

catalysts.

* Inhibitor $\rightarrow$ decreases the effectivity of catalysts

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Differential equation of SHO/SHM

Let a particle of mass m oscillate in simple harmonic motion. Now at time t it's displacement is x . Then

the velocity, $v = \frac{dx}{dt} \rightarrow (i)$, acceleration $= \frac{d^2x}{dt^2} \rightarrow (ii)$

$\therefore$ Now the magnitude of the force acting on the particle,

$$F = ma = m \frac{d^2x}{dt^2} \rightarrow (iii)$$

According to Hooke's law of vibration, force or acceleration is proportional to displacement and it is opposite direction

$$F \propto -x \Rightarrow F = -kx \rightarrow (v)$$

$$\therefore m \frac{d^2x}{dt^2} = -kx \rightarrow (vi)$$

$$\text{or, } \frac{d^2x}{dt^2} = -\frac{k}{m}x \rightarrow (vii)$$

Again, if the angular velocity of the particle is ω , then

we get, $\frac{d^2x}{dt^2} = -\omega^2 x \rightarrow (viii)$

Now from eq (vii) and (viii) we get

$$-\frac{k}{m}x = -\omega^2 x$$

$$\text{or, } \frac{k}{m} = \omega^2$$

Putting this value in eq (vii)

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

$$\Rightarrow \frac{d^2x}{dt^2} + \omega^2 x = 0 \rightarrow (ix)$$

Eq (ix) is the general differential equation of simple harmonic motion or oscillation.

By solving eq (ix) we get velocity and displacement

In order to solve eq (ix), let's multiply both sides by

$\frac{2dx}{dt}$ then we set,

$$2 \frac{dx}{dt} \cdot \frac{d^2x}{dt^2} + \omega^2 x \times \frac{2dx}{dt} = 0$$

Integrating the above equation

$$\left(\frac{dx}{dt}\right)^2 + \omega^2 x^2 = C \rightarrow (x)$$

Hence C is a constant of integration.

We need to find out the value of $x(t)$ if

Now when, $x = a$, then $\frac{dx}{dt} = 0$

When Inserting this condition in eq(x)

$$\omega^2 A^2 = C$$

$\therefore$ Now from eq(x) $\left(\frac{dx}{dt}\right)^2 + \omega^2 x^2 = \omega^2 A^2$

$$\text{or, } \left(\frac{dx}{dt}\right)^2 = \omega^2 (A^2 - x^2)$$

$$\text{or, } \boxed{\frac{dx}{dt} = \omega \sqrt{A^2 - x^2}}$$

Inserting this eq,

$$\sin^{-1}\left(\frac{x}{A}\right) = \omega t + \delta$$

Here δ is the constant of integration

$$x = A \sin(\omega t + \delta) \rightarrow (xi)$$

$$V_{\max} = ? \quad x=0$$

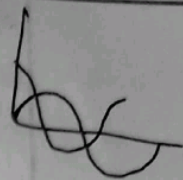
$$V_{\min} = ? \quad x=A$$

This is the general solution of ^{differential equation of} simple harmonic oscillation.

Next day $\rightarrow T$

(i) When $t=0$ then $x = A \sin \delta$

So, δ is the epoch or initial phase of the particle.



Now, if $\delta=0$ then from equation (i)

We get, $x = A \sin(\omega t + 0) = A \sin \omega t$

Here if $t=0$, then $x=0$, the motion of the particle starts from the mean position or equilibrium position.

(ii) Again, when $\delta = 90^\circ$ then from eq. (i) we get,

$$x = A \sin(\omega t + 90^\circ) = A \cos \omega t \rightarrow (12)$$

Here, if $t=0$, then $x=A$, in this case motion of the particle starts from the end or maximum point for other phases, initial displacement will be different.

a) Time period:

$$x = A \sin(\omega t + \delta) \rightarrow (13)$$

If time increased $\frac{2\pi}{\omega}$ we get

$$x = A \sin \left\{ \left(t + \frac{2\pi}{\omega} \right) \omega + \delta \right\}$$

$$= A \sin \{ \omega t + 2\pi + \delta \}$$

$$= A \sin (2\pi + \omega t + \delta)$$

$$= A \sin (\omega t + \delta)$$

So, it is seen that after time $\frac{2\pi}{\omega}$ the displacement of the particle become same, So, $\frac{2\pi}{\omega}$ is the time period of simple harmonic motion/oscillation.

$$\therefore T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\frac{k}{m}}} = \left[\frac{k}{m} = \omega^2 \right] \\ = 2\pi \sqrt{\frac{m}{k}} \quad (14)$$

Equation (14) is the equation of time period.

Eq (7)

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x$$

$$\Rightarrow \frac{m}{k} = \frac{-x}{\frac{d^2x}{dt^2}}$$

$$T = 2\pi \sqrt{\frac{-x}{(d^2x/dt^2)}}$$

$$T = 2\pi \sqrt{\frac{-\text{displacement}}{\text{acceleration}}}$$

This is the relation between displacement and acceleration.

(b) Frequency ,

$$n = \frac{1}{T} = \frac{1}{\frac{2\pi}{\omega}} = \frac{\omega}{2\pi}$$
$$= \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

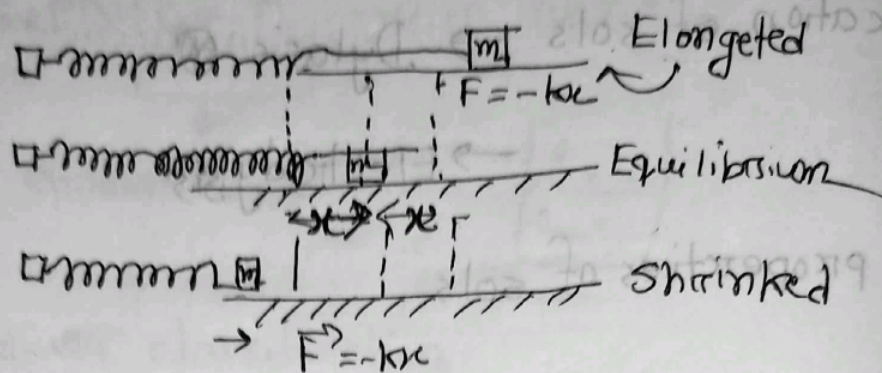
(c) Angular frequency :

$$\omega = \frac{2\pi}{T} = 2\pi n$$

$$= 2\pi \times \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$\therefore \omega = \sqrt{\frac{k}{m}}$$

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Potential, kinetic energy and total energy:

$$F \propto -x$$

$$\therefore F = -kx \rightarrow (i)$$

If the spring is released after pulling through a small distance it will oscillate in simple harmonic motion. If $\frac{d^2x}{dt^2}$ is the acceleration for displacement then,

$$F = ma = m \frac{d^2x}{dt^2} = -kx \rightarrow (ii)$$

$$\therefore m \frac{d^2x}{dt^2} = -kx$$

$$\therefore \frac{d^2x}{dt^2} = -\frac{k}{m}x \rightarrow (iii)$$

$$\text{Let } \omega = \sqrt{\frac{k}{m}} \text{ or } k = m\omega^2$$

$$\therefore \frac{d^2x}{dt^2} = -\omega^2x \rightarrow (iv)$$

$$\therefore \frac{d^2x}{dt^2} + \omega^2x = 0 \rightarrow (v)$$

equation of

This equation is similar to a simple harmonic motion.

So, motion of a vibrating spring is simple harmonic motion.

The solution of eq (v) becomes,

$$x = A \sin(\omega t + \delta)$$

Instantaneous velocity of the object,

$$v = \frac{dx}{dt} = A\omega \cos(\omega t + \delta)$$

$$= A\omega \sqrt{\cos^2(\omega t + \delta)}$$

$$= A\omega \sqrt{1 - \sin^2(\omega t + \delta)}$$

$$\Rightarrow v = A\omega \sqrt{1 - \left(\frac{x}{A}\right)^2}$$

$$\Rightarrow v = A\omega \frac{\sqrt{A^2 - x^2}}{\sqrt{A^2}} = \frac{A\omega \sqrt{A^2 - x^2}}{A} \rightarrow (vi)$$

$$v = \omega \sqrt{A^2 - x^2}$$

kinetic energy:

$$K = \frac{1}{2} m v^2$$

$$= \frac{1}{2} m \left(\frac{\sqrt{A^2 - x^2}}{A} \right)^2 \omega^2 \cdot (\sqrt{A^2 - x^2})^2$$

$$= \frac{1}{2} m \omega^2 (A^2 - x^2)$$

For very small distance displacement dx ,
the work done against restoring force $= F dx$

$\therefore$ Total work done for displacement x ,

$$W = -\int_0^x F dx = \int_0^x kx dx = \frac{1}{2} kx^2$$

Potential energy,

$$P.E = \frac{1}{2} kx^2 = \frac{1}{2} m\omega^2 x^2 \quad [\because k = m\omega^2]$$

$$= \frac{1}{2} k A^2 \sin^2(\omega t + \delta)$$

$$~~\frac{1}{2} k A^2 \sin^2~~$$

$$= \frac{1}{2} m\omega^2 A^2 \sin^2(\omega t + \delta)$$

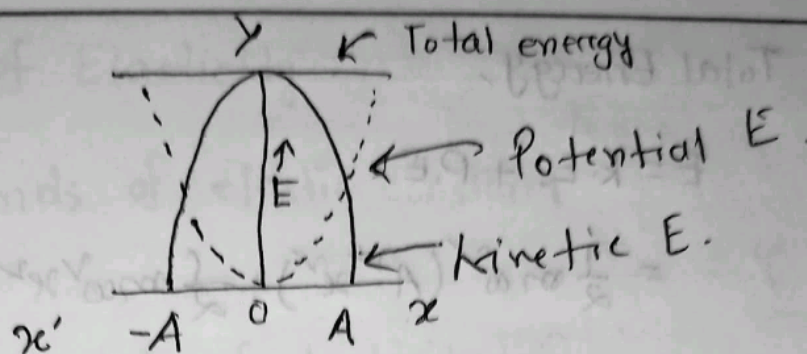
Total energy = K.E. + P.E.

$$= \frac{1}{2} m\omega^2 A^2 \cos^2(\omega t + \delta)$$

$$+ \frac{1}{2} m\omega^2 A^2 \sin^2(\omega t + \delta)$$

$$= \frac{1}{2} m\omega^2 A^2 \{ \cos^2(\omega t + \delta) + \sin^2(\omega t + \delta) \}$$

$$= \frac{1}{2} k A^2$$



- Fig: Variation of K.E., P.E.

and total energy with displacement,

Total Energy,

$$E = \frac{1}{2} k A^2$$

$$= \left(\frac{1}{2} k\right) A^2$$

$$\therefore E \propto A^2$$

Energy is proportional to the square of amplitude.

Total Energy,

$$E = K.E. + P.E.$$

$$= \frac{1}{2} m \omega^2 (A^2 - x^2) + \frac{1}{2} m \omega^2 x^2$$

$$= \frac{1}{2} m \omega^2 A^2 - \frac{1}{2} m \omega^2 x^2 + \frac{1}{2} m \omega^2 x^2$$

$$\therefore E = \frac{1}{2} m \omega^2 A^2$$

At the mean position $x = 0$

$$E = \frac{1}{2} m \omega^2 A^2$$

at end $x = A$

$$\text{Total energy } E = \frac{1}{2} m \omega^2 A^2$$

Total energy of a particle does not depend on ^{its} displacement and it remains same throughout its motion.

<u>S.I. unit</u>	<u>$2\pi \left(\frac{1}{x_1} + \frac{1}{x_2} \right)$</u>	<u>Moment of Inertia</u>
$T =$	$\downarrow$	$I =$
$V_{\max} =$	(i) $T = (=)$	
$V_{\min} =$		